CONCRETE BLOCK MASONRY

MES-ST-SPC-0007

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1. INTRODUCTION

1.1 Scope

This document provides the engineering requirements for plain and reinforced concrete block masonry for Ma'aden Projects.

This document is intended to be used in conjunction with the applicable documents listed herein.

1.2 Purpose

The purpose of this specification is to provide the requirements for the supply and testing of all materials, setting out, laying, reinforcing, finishing and inspection of plain and reinforced concrete block masonry for Ma'aden Projects.

1.3 Conflicts and Deviations

Any deviation from any of the requirements stipulated in this document shall go thru a waiver approval process as per Ma'aden Capital Investment System (MCIS) guidelines and shall be addressed to Ma'aden's Project Development & Engineering (PDE) team for their review and approval.

2. KEY TERMS AND DEFINITIONS

Employer	Saudi Arabian Mining Company (Ma'aden) and its authorized representatives and respective affiliates
Contractor	Party responsible for supplying materials, plant, labor and supervisory services
Manufacturer	Party who produces and warrants performance of products, materials, and/or items provided in accordance with contract documents

3. REFERENCED DOCUMENTS

This specification requires compliance with the provisions of the latest revision of the relevant standards and codes of the following authorities and technical organizations, as applicable, and specifications which shall be considered to form part of this specification, where applicable. In general, the Contractor shall be responsible for full compliance with applicable Ma'aden Health, Safety and Environmental Standards (MSHEMs). Any deviation from the MSHEMs shall require written approval from the Employer.

3.1 Statutory Regulations

All work performed and materials furnished shall comply with the applicable statutory regulations, codes and other requirements of the Kingdom of Saudi Arabia or local requirements of Project Site. The Contractor shall ensure that he is fully aware and informed of all Government laws, local ordinances, regulations and all orders and decrees of bodies or tribunals having any jurisdiction or authority over the Project and/or Site, which in any manner affect the Project and those engaged or employed on the Project, or which in any way affect the conduct of the Work. The Contractor shall at all times observe and comply with all such Government and/or Ministry laws, bylaws, ordinances, codes, regulations, orders and decrees.

3.2 Industry Codes and Standards

All work shall conform to the following applicable codes and standards.

3.2.1 *Local Codes and Standards (for Saudi Arabia)*

3.2.1.1. Saudi Building Code

All Ma'aden facilities shall comply with latest version of all Saudi Arabia applicable local codes and standards including but not limited to:

SBC 305	Structural – Masonry Structures

For project locations other than Saudi Arabia, the Contractor shall follow the respective applicable local regulations, codes and standards

3.2.2 *International Codes and Standards*

3.2.2.1. American Society for Testing and Materials (ASTM)

ASTM C33/C33M	Standard Specification for Concrete Aggregates
ASTM C73	Standard Specification for Calcium Silicate Brick (Sand Lime Brick)
ASTM C90	Standard Specification for Loadbearing Concrete Masonry Units
ASTM C129	Standard Specification for Nonloadbearing Concrete Masonry Units
ASTM C140/C140M	Standard Test Methods for Sampling and Testing Concrete Masonry Units and Related Units

ASTM C50/C150M	Standard Specification for Portland Cement
ASTM C207	Standard Specification for Hydrated Lime for Masonry Purposes
ASTM C270 REV A	Standard Specification for Mortar Unit Masonry
ASTM C476	Standard Specification for Grout for Masonry
ASTM C744	Standard Specification for Prefaced Concrete and Calcium Silicate Masonry Unit
ASTM C780	Standard Test Methods for Preconstruction and Construction Evaluation of Mortars for Plain Reinforced Unit Masonry
ASTM C1072	Standard Test Methods for Measurement of Masonry Flexural Bond Strength
ASTM C1314	Standard Test Method for Compressive Strength of Masonry Prisms
ASTM A615/A615M	Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement
ASTM A1064/A1064M	Standard Specification for Carbon-Steel wire and Welded wire Reinforcement, Plain and Deformed, for Concrete

3.2.2.2. American Concrete Institute

ACI PRC-524	Guide to Portland Cement- Based Plaster
ACI CODE-318	Building Code Requirements for Structural Concrete

3.2.2.3. The Masonry Society (TMS)

TMS 402/602	Building Code Requirements and Specification for Masonry Structures

3.2.3 ***Alternative Codes***

Alternative codes may be used for masonry works if approved by the Employer.

3.3 Ma'aden Standards

MD-101-SMEM-EG-ST-SPC-0003	Grouts
MD-101-SMEM-EG-ST-SPC-0005	Concrete Works
MD-101-SMEM-EG-ST-SPC-0006	Steel Reinforcing Bars and Wire Fabric.

3.4 Site Data

All materials, equipment and work methods shall be suitable for the Site conditions as defined in the applicable Site Data Specification.

4. MATERIALS

4.1 Concrete Blocks

Concrete masonry units shall conform to the following standards:

- ASTM C 90 for loadbearing concrete masonry units, or
- ASTM C 744 for prefaced concrete and calcium silicate masonry units.

The minimum design strength for concrete masonry shall conform to ASTM C90 Table 2.

Calcium silicate blocks shall be hard, durable, true to shape and size, and free from flaws, cracks and other defects. Blocks with broken edges or otherwise damaged shall not be used. They shall be protected from the harmful effects of direct sunlight and wind during curing and kept continuously moist for a period of at least seven (7) days. Blocks shall not be used in the works until twenty-eight (28) days have elapsed since their date of manufacture.

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4.2 Mortar

All mortar, materials used in mortar and testing of mortars shall conform to ASTM C270. All sand, cement and lime shall be measured in gauge boxes before being packed in the mixer. Dry ingredients shall be placed in the mixer and thoroughly blended before the addition of water and lime or admixtures. Mixing shall continue until all ingredients are thoroughly mixed but not for less than three minutes. Hand mixing shall not be permitted. Mortar shall not be used after the initial set has taken place and re-tempering shall not be permitted. Mortar shall be used within 30 minutes of the mixing. Premix mortars may be used conforming to the requirements of SBC 305.

Mortar for use in masonry construction shall have the minimum design strength as shown in SBC 305 Table 3.7.2, depending on mortar type. Generally, the mortar Type M shall be used with average compressive strength of 17 MPa at 28 days.

The mortar proportions shall conform to SBC 305 Table 3.7.1 and the acceptable aggregate material shall be fine engineered sand. Sand up to 0.063 mm shall not be more than 8% by weight of the total quantity.

The following requirements for mortar shall apply:

- The proposed mix design, samples of material and test cylinders or prisms shall be submitted to the Employer for approval and testing
- The Employer may request test cylinders or prisms to be taken from random mortar batches throughout the work
- In the event of any test specimens failing strength tests the Employer shall determine the action to be taken
- The mortar generally shall comply with ASTM C270 although mixes using admixtures in place of lime may be used if approved and provided tests show that the minimum mortar strength is attained
- Cement used for over ground works shall comply to ASTM C150/C150M
- Cement used for underground works shall be sulphate resistant.

4.3 Admixtures

Where approved by the Employer, an admixture may be used as a plasticizing and densifying agent provided the minimum strength is maintained.

4.4 Concrete Grout

Grout shall conform to SBC 305 Section 3.10.0 and ASTM C476.

Note: Term 'Concrete' is interchangeable with term 'Concrete Grout' when used as a filler in masonry cores.

4.5 Reinforcement

All reinforcement shall conform to SBC 305 Section 3.11 and 3.12 and ASTM A615/A615M, ASTM A1064/A1064M correspondingly.

4.6 Metal Lath

Metal lath shall be supplied in widths 40 mm less than the nominal width of the block work.

4.7 Flashing

Cavity flashings and apron furnishings shall be galvanized sheet metal with a minimum thickness of 0.8 mm.

4.8 Caulking Materials

Caulking material shall be suitable approved proprietary skinning type caulking compound colored to match blocks.

5. WORKMANSHIP

5.1 General

All work shall be carried out using storey rods marked to conform with course dimensions specified.

Block work shall be set out accurately, plumb, level and properly bonded, with walls carried up evenly, no part rising more than four courses above any other point. Rising work shall be raked back and not toothed. Jambs, reveals, corners, etc shall be true and plumb and perpends shall be true to line.

The erection of concrete masonry shall be suspended if in the opinion of the Employer prevailing climatic conditions or inadequate facilities furnished by the Contractor prevent proper bonding, setting or curing of mortar joints.

Items required to be built in with masonry, including anchors, wall plugs, door frames and accessories, shall be built in as erection progresses. Any small variances which occur shall appear on the interior surfaces of exterior walls or on the room side of corridor walls.

5.2 Gauge

All block work shall be built to a gauge rod of 100 mm or 200 mm, dependent on block size specified including the 10 mm joint.

5.3 Bond

All block work shall be built in stretcher bond unless a stack bond is shown on the drawings or specified.

Where existing work is extended a straight control joint shall be formed and existing and new block work tied together with either dowels set into holes drilled into the bed courses or straps secured in an approved manner to the existing walls and built into each bed course of the new wall.

5.4 Joints

Unless otherwise specified or shown, all horizontal and vertical joints shall be 10 mm in width with a tolerance of ± 3 mm where extensions are made to existing metric size block work, particular care shall be taken to match existing joint widths and lines.

Solid blocks shall have full bedding and filled vertical joints. In reinforced and pre-stressed block work horizontal and vertical joints shall have full mortar coverage on all face sheets, webs and ends with a maximum intrusion into the core space of 5 mm. Horizontal joints in plain block work comprising hollow units shall be buttered only for the thickness of the face shell. Lintels, capping units and all bearing plates shall be set on a full bed of mortar. Any blocks which are disturbed after mortar has stiffened shall be removed.

Before block laying commences the top surface of concrete foundations, beams, floor slabs, etc shall be thoroughly cleaned to ensure good bonding. The bed joints shall be laid with a full mortar coverage except that areas where core filling concrete is required shall be kept free of mortar so that core filling concrete, when placed, shall be in direct contact with the cleaned foundation concrete.

Finish for joints shall be as follows:

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- All blocks shall have perpends solidly filled and the joints shall be finished, raked and jointed with a 10 mm recess. Polishing of joints when mortar is dry will not be permitted; darkening of joints so caused will require a relaying of areas affected
- Where covered or ceramic tile skirting is to be applied, mortar joints at the base of all internal walls shall be tool flushed.

Control joints shall be located at a maximum spacing of 15 m throughout concrete masonry walls and shall be constructed as shown on the drawings.

5.5 Cavity Walls

Cavity walls shall have a 50 mm nominal width cavity or such other width as shown on the drawings.

Galvanized wire ties shall be built in at 400 mm centers vertically, staggered and set not more than 600 mm apart nor more than 300 mm from any opening or more than 300 mm apart around the perimeter of any opening; at least one tie for every 0.3 m2. Weep holes shall be formed by starting at the second joint in from the ends and leaving every forth vertical joint of the bottom course of the wall open. Similar weep holes shall be formed wherever through flashings occur

Battens shall be used to protect wire ties from mortar droppings. Openings shall be left in walls in positions directed and on completion an inspection of the cavity shall be made with lights. Should cavities be found to be in any way obstructed, the Contractor shall clean out all such obstructions to the approval of the Employer.

Cavities shall be filled to a height of 75 mm above adjacent finished ground level with 1:3:6 (cement:sand:aggregates) concrete, weathered at top and finished at a horizontal joint in the outer leaf.

5.6 Face Blockwork

All face block work shall be completely protected during the Works.

Before placing concrete, all face block work shall be completely covered with waterproof sheathing in stretcher bond and set out to avoid cutting blocks.

All face block work shall be extended two courses below natural ground line or one course below external paving unless otherwise detailed.

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Mortar jointing in each wall plane shall be of an even color. To achieve this when different mortar mixes are used they shall be color matched to the approval of the Employer by the substitution of white for grey cement or other means.

Block work in ducts, behind built-in fittings and in plant rooms shall be built in facework and finished smooth and true.

All external block work shall be face blocks unless otherwise specified. The locations of face blocks shall be as indicated on the drawings.

The extent of the internal face block work shall be as indicated on the Schedule of Finishes.

5.7 Sills

Window sills shall be formed with standard nominal 200 x 100 solid blocks laid in soldier course along full width of window.

5.8 Damp Proof Courses

Damp proof courses shall be formed with Super Grade Alcor or other approved equivalent, laid in continuous lengths dry, for the full width of blocks and lapped 100 mm minimum and as specified by manufacturer at all joints and corners.

Damp proof courses shall project 40 mm from internal face of block leaves where concrete floor slabs on ground are indicated. Damp proof course shall be neatly dressed down and secured to floor slab over membrane which shall be turned up against all walls.

Damp courses to leaves of all external block cavity walls and all internal block walls shall be installed within the thickness of the ground floor construction.

5.9 Flashing

All necessary flashings shall be built in Super Grade Alcor or approved equivalent to discharge water from outer leaf and to ensure watertight construction.

Flashings shall be secured and sealed to frames in continuous lengths, where jointing is unavoidable at corners and in long lengths, lap at least 150 mm and seal with mastic. Flashings shall be a minimum of 150 mm wide at jambs and 225 mm wide at heads and sills of openings.

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Wherever construction in exposed locations is interrupted horizontally, cavities shall be flashed with Super Grade Alcor, built into inner leaf, stepped down two courses minimum and built into horizontal joint of outer leaf.

Weep holes shall be formed by leaving open perpends at maximum 1.3 m centers.

5.10 Junction of Walls

Where non load bearing walls intersect, every second course shall be tied in with strips of galvanized reinforcement.

Where walls intersect with at least one being load bearing tie in every meter vertically with 700 mm x 30 mm x 6 mm galvanized mild steel ties with 50 mm right angle bend upward at each end and encase in concrete with metal lath as formwork in accordance with the manufacturer's instructions.

5.11 Horizontal Strip Reinforcement

Horizontal strip reinforcement shall be placed in the bed joints of plain blockwork as shown on the drawings. The minimum requirements shall be one layer of mesh of the specified type and dimensions placed in each of the first and second bed joints, joints immediately above and below openings and in alternate bed joints throughout the rest of the wall. Strip reinforcement shall be extended 1 m either side of openings. Strip reinforcement shall be carefully laid into each mortar bed with no part of the reinforcement less than 10 mm from the block faces. Splices shall have a minimum overlap of the end cross wires in each strip equal to the cross wire spacing plus 25 mm.

5.12 Reinforced Concrete Masonry

Masonry walls shall be laid up to the full height before cores are concreted. Horizontal reinforcement with laps, splices and bends shall be fixed in web grooves in the top of blocks as the blocks are laid.

Vertical reinforcement shall be introduced from the top after clean out and securely tied to the bottom starter bars or cross rods.

All walls in which cores are to be filled with concrete shall be constructed to preserve the unobstructed vertical continuity of the cores. Cores to be filled shall have an unobstructed continuous vertically aligned core of not less than 75 mm by 100 mm. At the base of cores to be concrete filled and where reinforcement is spliced, clean out openings shall be provided using open face blocks.

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Debris and overhanging mortar shall be removed from the cores before vertical reinforcement is introduced and tied. After inspection and approval the clean out openings shall be sealed by buttering on the facing units, and shored up ready for pouring concrete.

Core fillings shall be placed and compacted to the full height of a wall or column in lifts not exceeding 3 m in height. The concrete shall be rodded or puddled to ensure complete filling of the core. Where pours are broken or to be continued after additional block laying, concrete shall be kept 25 mm below the top of the top block. The surface shall be scrabbled back to expose aggregate and cleaned before any further concrete is poured. Metal lath shall be used to prevent concrete entering cores to be left unfilled and to confine concrete to horizontal bond beams.

Placing of concrete in bond beams and lintels shall be completed in one continuous pour.

5.13 Plain Block work Below Grade

Cores of concrete blocks for plain block work laid below shall be completely concrete filled up to one course above grade.

5.14 Curing

All concrete masonry block work shall be kept dry and protected from adverse weather conditions. When the atmosphere is exceptionally dry, and when directed by the Employer, the surfaces of the wall shall be kept moistened by means of a fog spray for a period of up to three days after laying.

5.15 Cutting

When cutting of blocks is in the Employer's opinion unavoidable, it shall be done using a masonry saw. Cuts shall be straight and perpendicular to the plane of the exposed face. Cutting and patching of masonry required to accommodate the work of others shall be performed using a masonry saw and/or power tool technique designed for the purpose.

5.16 Unfinished Work

Unfinished work shall be stepped back for jointing with new work. Before new work is commenced, all loose mortar shall be removed and the exposed joints thoroughly cleaned.

5.17 Built In Work

All structural ties, anchors, chases, frames and sleeves and provisions for all other work shall be built in as block laying progresses. Blocks shall be carefully and accurately cut as specified for the

installation of tile accessories, electrical outlets and conduits, plumbing lines and mechanical installations. Pressed metal door frames shall be filled solid with mortar as work proceeds.

5.18 Cleaning

Extreme care shall be taken to prevent damage to mortar and concrete splotches on exposed surfaces. All forms shall be made tight and concrete spilled onto surfaces shall be removed before it sets. By the completion of the Works, all exposed masonry surfaces shall be cleaned down and all holes in exposed joints filled with mortar to leave masonry clean, free from mortar and loose concrete. All scaffolding equipment, surplus material and debris shall be removed from the unfinished work. No acids shall be used in the cleaning of block masonry.

5.19 Plastering works

Plastering work shall conform with ACI PRC-524, Guide to Portland Cement based plaster.

Portland Cement plaster shall be applied to concrete masonry surfaces in two or three coats, consisting of basecoat(s) and a finish coat. Prior to application of the basecoat, uniformly pre-moisten concrete masonry surfaces to aid in curing of the plaster. However, avoid saturating the masonry either by excessive spraying or by exposure of uncovered masonry walls or units to rain.

Concrete masonry walls should be properly aligned before plastering minor deviations from true plane; however, they can be fixed using leveling coats of plaster. Excessive deviation of the surface plane or misalignment may require furring and lathing to restore true plane. Concrete masonry units that are chipped, broken, or out of line should be patched and brought level to the plane of the substrate surface.

5.20 Protection of Work

The top of the unfinished masonry work shall be covered to protect it from weather.

Hot and cold weather protection shall be applied on masonry based on the SBC 305, Section 4.3 and 4.4.

5.21 Expansion Joint

Expansion joints shall be placed in masonry walls with continuous lengths of 15 m or more. Where the spacing of expansion joints is less than or equal to 10 m, the expansion joint shall have a width of 15 mm, otherwise it shall have a width of 20 mm.

Expansion joints shall be closed by either:

a) A compressible foam or polystyrene material and finished with a gunned sealant, or
b) A metal joint insert.

5.22 Tolerance

Masonry, except masonry veneer, shall be constructed within the tolerances specified in ASCE 6/TMS 402/602.

5.23 Accessories

Accessories for use in concrete block work are listed in Table 5.1-

Table 5-1 : Accessories for Use in Concrete Block Work

Accessories	Locations to be used (Location of Application)
Expanded Metal	• Fixed to walls interfaces with ceilings, steel beams and columns for application of fire-proof sprays or plastering.
Plaster Stop Beads	• Used at corners of wall of surfaces to terminate wall plaster or render finish • Terminate plaster edge at doors and windows.
Block Reinforcement	• Used in the middle of the block (vertical or horizontal) • Horizontal reinforcement placed in web grooves in the top of blocks as the block is laid (at least one 12 diameter bar with not more than 1200 mm spacing) • Vertical reinforcement required to be securely tied at bottom to starter bar reinforcement. (at least one12 diameter bar with not more than1200 mm spacing) • Steel wire for horizontal and vertical reinforcement shall comply with SBC 305.

Accessories	Locations to be used (Location of Application)
Laths	• Base for plaster ceilings, ceramic wall ties on timber frames, soffits and other architectural effects • Support fireproofing spray on steel columns and beams in fire rated construction and for column rendering.
Meshes	• Between bricks course for bonding internal and external leaf • Plastering base.
Facing	• Interior and external wall architectural features.
Cavity Ties	• Connect masonry to masonry in cavity walls • Connect masonry to timber frame.
Dovetail Anchors	• Used to fix window/door frame to masonry • Anchor masonry to concrete.
Lintel	• A beam that supports the wall above windows, doors or other opening.
Wire Ties	• Connect masonry veneer wall cladding to a timber or metal load-bearing frame • Connect two leaves of cavity masonry • Connect two leaves of masonry forming a single wall with no cavity • Connect intersecting masonry walls at a joint where course bonding is not employed.
Horizontal strips	• Horizontal strip reinforcement shall be placed in the bed joints of plain block work, similar to mesh.

6. INSPECTION

All block work shall be inspected visually. All units shall be sound and free of cracks or other defects that would interfere with proper placing of the unit. All defects observed on the surface of the

concrete block wall shall be referred to the Employer and repaired as instructed or by an approved method.

Testing of mortars, and samples/specimen requirements, to comply to ASTM C270, ASTM C780 and ASTM C1072

Testing of Masonry units for compressive strength, and samples/specimen requirements to comply to ASTM C1314

All submittals (material test reports, mortar mix design, etc.) to be submitted to the Employer for review and acceptance within 7 days for test completion.

Generally the quality assurance shall conform to SBC 305 Chapter 3 and 5.